

BOM

Component	Mass g/cell	kg/kWh	Formula / source
Positive active material (NMC811)	16.168	0.7418	pos coating 16.842 g x 0.96 AM mass fraction (literature estimate); coating = cell/r5_v11_energy.json:layer_kg_m2 positive 0.163994 x area 0.1027 m2
Positive conductive additive (carbon black)	0.337	0.0155	pos coating x 0.02 (literature estimate)
Positive binder (PVDF)	0.337	0.0155	pos coating x 0.02 (literature estimate)
Positive coating total (set caliber)	16.842	0.7727	parameter set: 75.6um x (1-0.335) x 3262 kg/m3 x 0.1027 m2 = layer_kg_m2 x area
Negative active material (graphite)	11.961	0.5487	neg coating 12.525 g x 0.955 (literature estimate); coating = layer_kg_m2 negative 0.121955 x 0.1027 m2
Negative binder (CMC)	0.188	0.0086	neg coating x 0.015 (literature estimate)
Negative binder (SBR)	0.376	0.0173	neg coating x 0.030 (literature estimate)
Negative coating total (set caliber)	12.525	0.5746	parameter set: 115um x (1-0.36) x 1657 kg/m3 x 0.1027 m2
Separator (polyolefin)	0.259	0.0119	layer_kg_m2 separator 0.00252492 x 0.1027 m2
Electrolyte (EC/EMC + LiPF6)	8.918	0.4091	pore volume 7.432 cm3 x 1.2 g/cm3 (literature density) x fill factor 1.0; pores = (75.6e-6*0.335 + 12e-6*0.47 + 115e-6*0.36) x 0.1027 m3
Positive current collector (Al foil)	4.437	0.2036	layer_kg_m2 pos cc 0.0432 x 0.1027 (16um x 2700 kg/m3)
Negative current collector (Cu foil)	11.042	0.5066	layer_kg_m2 neg cc 0.10752 x 0.1027 (12um x 8960 kg/m3)
Enclosure, tabs, terminals	Not modeled	Not modeled	no parameter in set
Total (contract caliber, electrolyte excluded)	45.105	2.0693	cell/r5_v11_energy.json:mass_kg = 0.0451052 kg; energy = 21.7970 Wh
Total (with electrolyte estimate)	54.023	2.4785	contract mass + electrolyte row above (estimate, not adjudicated)